CAI6734 Week 11 Individual Progress Report

Testing in Imagination:
World-Model Planning for Language Agents
in Sequential Clinical Diagnosis

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This report covers my individual work for Week 11 (November 4–10, 2026), from records through November 8. I report no November 9 or 10 activity. The credentialed run and the finalization checks in scope are finished. No new training was started during finalization. The numbers are a replication attempt, not a reproduction of the paper, and matching the original findings remains unresolved. No PHI. Week 10 synthetic checks are not these results.

1 Progress this week

1.1 Access, storage, validation, and the split

- Access to MIMIC-IV-Ext CDM v1.1 opened on November 4. On November 5 I downloaded the release and verified all 12 listed SHA256 digests, line by line. PhysioNet's `SHA256SUMS.txt` uses a single space, so macOS `shasum -c` can reject a line and still exit 0.
- Both copies are owner-only and outside every git repository: a private Mac directory that is not cloud-synced, and private HiPerGator storage the class group cannot enter. The Downloads copy was deleted on November 5 after a byte-for-byte comparison. No data file is in git. `check-git` printed `tracked-file check: PASS` before the related commits, including `b546c2e` and `2478dae`.
- v1.1 ships CSVs; the pinned splitter reads four condition pickles. The pickle builder follows Hager et al.'s `ConvertPhysionet.py` (`4d33f98`). Outputs were byte-identical on the Mac (arm64) and on HiPerGator (x86_64). `validate` passed on both. The cohort is 2,400 admissions, classes 957/648/257/538. The lab mapping has 1,209 rows (118 blank `itemid` groups). Labs are 138,788 rows plus 4,403 microbiology rows (143,191). Radiology has 5,960 reports where upstream states 5,959; that gap is unexplained. 373 reports use 10 modalities the upstream parser skips.
- Seed 269 is the pinned splitter's seed. D-4 still says there is no canonical seed; I used 269 for parity and recorded the discrepancy. The split is 1,920/240/240, test classes 96/65/25/54, with no admission shared across train, validation, and test. v1.1 has no `subject_id`, so subject leakage is unassessable and the subject-grouped split cannot run. I do not call that overlap zero. The S6 leakage audit says the same.

1.2 Runtime, deviations, and the finished run

- HiPerGator has no A100s. The class QOS shares 32 CPUs, 250 GB, and 8 GPUs; the burst QOS has no GPUs. Pinned torch 2.6.0+cu124 stops at `sm_90` (job 49867089). The Blackwell

runtime keeps upstream code at `3f435a10` and changes 28 pins (torch 2.7.0+cu128, vLLM 0.9.2, TRL 0.15.2, transformers 4.51.1). Adapters are evaluated as merged weights, as upstream does, because the vLLM LoRA kernel fails on `sm_120` (job 49868789; fix `d434552`).

- D-1 keeps incorrect reward 0 as primary and puts −1 only in the named ablation. D-3 and D-5 use `nvidia/Llama-3.1-Nemotron-Nano-8B-v1` at `54641c16` as summarizer and policy. The ledger records that as the authorized HiPerGator default: the paper's Qwen backbone is not permitted. Stock Llama-3.1-8B-Instruct is the fallback ablation, not a model the written criteria require, and it was not trained. D-6 is greedy evaluation; upstream samples at temperature 1.0.
- Graph `7fcd9e5` finished. S0 passed at 4:04 PM EST on November 6. All 22 jobs were `COMPLETED 0:0`. S6 finished at 5:10 AM EST on November 8 (receipts `f323a460`, `05601581`, `a03cb630`, `800c18a6`). Four arms trained 3,840 steps, about 25–28 h each, then the 240 test admissions were scored. Graphs `b7190c5` and `4d22c70` were cancelled for the failures below and were not scored.

1.3 Primary results (greedy) and the separate decoding check

Finalization did not re-run evaluation, S5, or S6. `metrics.json` is the S6 export. Intervals are 95% class-stratified patient bootstraps (10,000 resamples) on one training seed, 269. They cover test-admission sampling only, not between-seed or subject-cluster uncertainty. A paper value inside an interval is a description of that comparison, not equivalence and not a reproduction.

ID	Metric	Paper	Observed	95% interval
T1	Appendicitis recall (%)	93.1	90.62	[84.38, 95.83]
T2	Cholecystitis recall (%)	83.6	72.31	[61.54, 83.08]
T3	Diverticulitis recall (%)	75.0	72.00	[56.00, 88.00]
T4	Pancreatitis recall (%)	73.5	57.41	[44.44, 70.37]
T5	Mean of four recalls (%)	81.3	73.09	[66.65, 79.19]
T6	Micro F1 (%)	84.1	81.15	[76.47, 85.53]
T7	Macro F1 (%)	81.3	78.20	[72.32, 83.44]
T8	Provided-test cost (USD)	1,295.61	1,439.58	[1,341.50, 1,537.65]
B1	Zero-shot mean recall (%)	64.5	12.71	[8.04, 17.86]
B2	ReAct mean recall (%)	74.9	gap	no ReAct at the pin

T5 minus the paper is −8.21 [−14.65, −2.11]. The published points for T1, T3, T6, and T7 fall inside the intervals; T2, T4, T5, and B1 lie below; T8 lies above. Pooled accuracy is 76.25%. Hypothesis accuracy went from 29.59% to 81.67% [77.24, 85.89] (C1; paper 75.7 to 81.9). ECE went from 0.3022 to 0.1281 [0.0861, 0.1749] (C2): improved, still above the paper's 0.037. These are held-out endpoints. Validation never used test data: 31.21% before training, 75.15% at the final checkpoint.

C3, a lower cost from the cost reward, was not observed. The reward is present on all 3,840 primary steps (residual 0.0) but small, about −0.09 to −0.11 per rollout against about 0.17–0.53 for diagnosis. no_cost to primary was \$1,321.33 to \$1,439.58 and 70.81% to 73.09%, paired cost +\$118.25 [−0.25, 239.86], Holm $p = 0.551$. C4 was 57.49% to 73.09%, paired +15.59 [6.34, 24.60], Holm $p = 0.00056$; the paper's Table 3 number is absent, so only the direction is assessed. The D-1 ablation scored 59.38% and \$1,619.27. Greedy primary minus that arm is +13.71 [5.32, 21.93], Holm $p = 0.0031$, and that call is decoding-sensitive.

The audited zero-shot path matches upstream `3f435a10` on the prompt, chat template, regex parser, action names, stop string, and 256-token budget. The system message and greedy decoding are the D-5 and D-6 differences already named. No code bug was found on those paths. That is not proof that no bug exists. Greedy zero-shot left 194 of 240 episodes undiagnosed (23 with no valid test request, the only group that can be a first-action parse failure). The primary arm left

29, none of that kind. Episodes do not separate parse failures from the action budget or the token limit.

Temperature 1.0 is a post-hoc check on the same 240 admissions (jobs 50282940–50282944 and report 50282959; code `4b79683`, then `d3ef9de`). It does not replace the greedy table. Zero-shot mean recall was 9.19 [5.55, 13.44], paired −3.52 [−9.38, 2.37]. Primary mean recall was 59.30 [52.23, 66.19], paired −13.79 [−21.70, −6.24]. Sampled decoding did not close the gap. That is not proof that decoding is causally ruled out. The backbone and summarizer stay untested candidate explanations, not a demonstrated sole cause. At 1.0, C3 is still absent (cost +$82.94 [−30.33, 195.84], Holm $p = 0.538$) and the D-1 recall gap is +3.55 [−5.68, 12.76], Holm $p = 0.474$. C4 stays positive (+18.66 [9.94, 27.21], Holm $p = 0.000049$). An independent code review raised five warnings; all were fixed test-first in `d3ef9de`, where 264 tests passed (181 here, 83 in `lacdm/tests`). Only the three aggregate files are tracked (`2478dae`). Four old work directories were removed at 1:30 PM EST on November 8; `7fcd9e5` was kept.

2 Setbacks and how I addressed them

- **Validation would have crashed.** The best-metric setting expects `eval_loss`, which per-patient validation does not report. I cancelled jobs 49874617–49874620 first. Commit `4d22c70` drops that setting and validates once per epoch. One pass takes about 80 minutes, so the old schedule would have added about 50 GPU-hours per arm. Smoke job 49889947 passed, and every finished arm passed both epoch checks.
- **Out of memory at step 333** (`decision_only`, job 49889996): 12.31 GiB needed, 12.27 GiB free, 4.70 GiB fragmented. I resumed with expandable allocator segments (job 49924900) and then set that allocator in every stage (`7fcd9e5`).
- **Crash at step 1,340** in the three hypothesis arms (jobs 49889995, 49889997, 49889998). Upstream runs `str.format` on the whole episode, and that record contains braces. A failing test led to a literal substitution in `7fcd9e5`. Smoke job 50010187 passed, and the finished run passed that step.
- **Class memory cap.** Other jobs held about 240–246 of 250 GB, so S4 waited. Host memory peaked at 6.8 GB; queued requests dropped from 40 GB to 16 GB. CPU-only S4–S6 moved to the burst QOS. GPU evaluation waited 6 h 40 min to 7 h 20 min, then finished in 24–33 minutes each.

3 Next steps

1. Keep paper matching unresolved. Full completion in the ledger is T5–T8 within an agreed tolerance, and no tolerance is stated, so that bar is unassessable. This run is not a reproduction.
2. Further GPU work is a separate decision, and neither option ran: about 105 GPU-hours for another seed, or about 110 for the stock Llama-3.1-8B ablation. That ablation would test one candidate. It would not reproduce the paper's Qwen and Mixtral stack. B2, subject leakage, and the 5,960-versus-5,959 radiology count stay open until that decision.

Notes

The 22 September 2026 definition of done says to pick one option. Full completion is T5–T8 within an agreed tolerance. No tolerance was ever stated, so that option is unassessable, and a published point inside an interval does not meet it. The 8 November status says the partial option is met and exceeded, because training did run, and the README sets no numerical match. Figures are aggregates only. One seed, D-3, D-5, the 28-pin runtime, and D-6 keep this a replication attempt.